

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2024

Programme: BE

Full Marks : 100

Course: Electronics Devices and Circuits (New)

Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

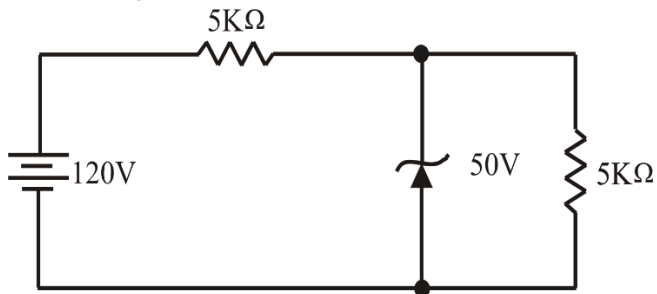
Attempt all the questions.

1. a) Explain the working principle of p-n junction diode under 7
- i. no voltage applied
 - ii. forward biasing
 - iii. reverse biasing

OR

What is semiconductor material. Explain different type of semiconductor.

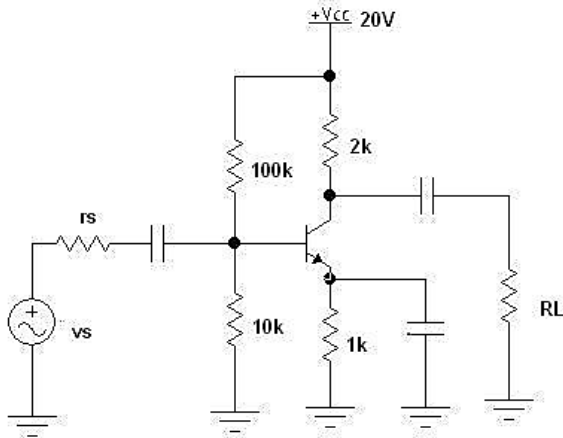
- b) Determine the following parameter for a given circuit diagram. 8
- i. The output voltage
 - ii. Voltage drop across the series resistance
 - iii. Current through the Zener diode



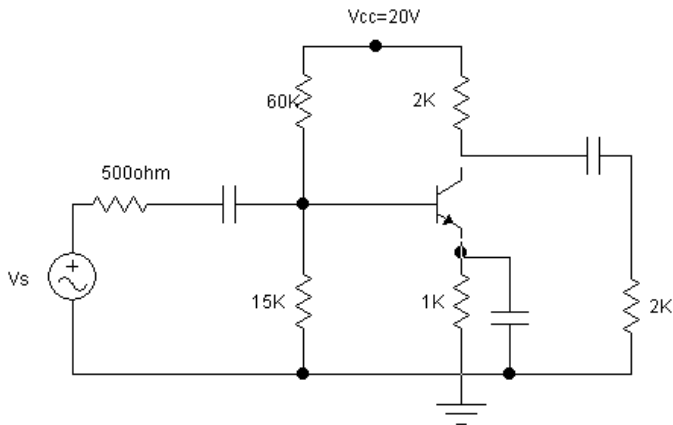
2. a) Draw the input and output characteristics curve for a common emitter configuration. Also, explain why CE configuration is used for amplification. 7
- b) Design a self-biasing circuit for a Si transistor with $V_{CEQ}=10V$, $I_{CQ}=5mA$ and $S=10$ with $\beta=150$ and $V_{CC}=40V$. 8

OR

For the BJT circuit shown below, draw dc load line and locate operating point on it. (assume $\beta=100$)



3. a) Explain the importance of power supply in electronic devices. Differentiate between regulated and unregulated power supplies. 7
- b) Explain the construction and working principle of MOSFET. Point out the difference between DMOSFET and EMOSFET. 8
4. a) What is multistage amplifier? Derive necessary expression for super beta transistor. 7
- b) For the given voltage divider circuit, find A_{vs} , A_v and A_i , given $\beta=100$ and $r_o=200K\Omega$ Draw re model circuit. 8



5. a) What do you mean by power amplifier? Explain the operation principle and derive efficiency of Class A amplifier with the help of circuit diagram. 7
- b) What is Barkhausen Criterion for oscillation? Design a RC phase shift oscillator to operate at a frequency of 8 kHz. 8
- OR**
- Explain Barkhausen criteria of oscillation? Design a Wein-bridge oscillator to operate at a frequency of 24 kHz.
6. a) Design an operational amplifier that produces an output equal to $V_o = -V_1 + 2V_2 - 3V_3 + V_4$. 7
- b) What is importance of feedback amplifier. Derive the closed loop gain of feedback amplifier. 8
7. Write short notes on: (**Any two**) 2×5
- a) Virtual ground
- b) Clipper and clamping circuits
- c) Bias stabilization